Singular Learning Theory for SubRep: Goal–Subgoal LLC Relations in Hierarchical Planning with Functional Coupling and Shared Representations

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December 10, 2025

Abstract

SubRep is a framework for compositional goal pursuit in which options export expectation models $(\hat{r}, \hat{n})$ and are admitted into a planner-facing library via cone-dominant (CDS) or Pareto-dominant (PDS) tests over a motive geometry. This paper develops a rigorous singular learning theory (SLT) analysis of SubRep systems, addressing the question: how do the local learning coefficients (LLCs) of a goal-level system relate to those of its subgoal modules?

We extend previous work in six directions. First, we introduce explicit *functional coupling terms* arising from planner–option composition and derive sufficient conditions under which such coupling remains dominated, preserving approximate LLC additivity. Second, we analyze *shared representation* architectures where options and planners share feature extractors, showing how the shared block's LLC contributes once (not per-module) to the total complexity. Third, we address the *non-analyticity* introduced by hard CDS/PDS admission tests by introducing soft relaxations and proving convergence of the relaxed LLC to a well-defined stratified limit. Fourth, we develop a comprehensive treatment of *non-convex motive geometries*, decomposing general motive spaces into convex cells, analyzing the tropical/piecewise-linear structure that emerges, and characterizing singularities at cell boundaries via a double stratification. Fifth, we analyze the *robustness of additivity*, proving quantitative bounds on the "additivity gap" as conditions degrade, and characterizing graceful versus catastrophic failure modes. Sixth, and most significantly, we move *beyond additivity* by introducing *LLC interaction complexity*—an analog of mutual information that captures how subgoal complexities combine non-additively. This leads to an inclusion-exclusion decomposition where pairwise, three-way, and higher-order interaction terms replace the naive additive model, with positive interactions indicating redundant complexity (shared singular structure) and negative interactions indicating synergistic complexity (emergent singularities from combination). Together, these results provide a complete framework for understanding goal–subgoal LLC relationships in hierarchical planning systems.

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1 Introduction

SubRep decomposes complex goals into subgoals (options) by combining expectation models $(\hat{r}, \hat{n})$ for options, a motive geometry (cone W or finite weight cover W_{ref}), cone-dominant (CDS) or Pareto-dominant (PDS) admission tests, and a Motive Decomposition Network (MDN) that co-learns motive factors and weight structure. The result is a library of subgoals with planner-facing certificates: each admitted option is guaranteed (up to margins and model error) to improve backed-up values on a motive slice.

Singular learning theory (SLT) characterizes the geometry of low-loss regions via local evidence integrals and the local learning coefficient (LLC) λ. In modular systems, understanding how λ decomposes across modules is central to analyzing learned representations and model complexity.

Previous treatments of LLC additivity in modular systems assume independence or weak coupling between modules. However, SubRep systems exhibit four features that complicate this picture:

1. **Functional coupling.** The planner module takes option outputs as inputs; option quality directly affects planner loss gradients.

2. **Shared representations.** Modern implementations often share feature extractors across options and planners.

3. **Discrete admission tests.** CDS and PDS involve hard thresholds that introduce non-analytic boundaries in the loss landscape.

4. **Non-convex motive geometries.** Real motive spaces arising from learned MDNs are typically non-convex, creating piecewise-linear structure in the margin functions and additional singularities at cell boundaries.

This paper addresses all four issues. Section 2 reviews SLT and establishes notation. Section 3 formalizes SubRep in loss-based language. Section 4 analyzes functional coupling and derives domination conditions. Section 5 treats shared representations. Section 6 introduces soft CDS/PDS and develops stratified SLT analysis. Section 7 develops the theory for non-convex motive geometries, including convex decomposition, tropical structure, and double stratification. Section 8 states our main decomposition theorems. Section 9 analyzes how additivity degrades as conditions weaken, characterizing graceful versus catastrophic failure modes. Section 10 moves beyond additivity entirely, introducing LLC interaction complexity as an analog of mutual information and deriving an inclusion-exclusion decomposition that treats non-additive effects as meaningful structure rather than error. Section 11 discusses implications for SubRep design.

2 SLT Background

We summarize the SLT machinery needed for our purposes, following Watanabe's foundational treatment.

2.1 Setup and notation

Let $\Theta \subset \mathbb{R}^d$ be a parameter space with prior density $\phi(\theta)$. Given data $D_n = \{x_1, \ldots, x_n\}$ drawn i.i.d. from true distribution $q(x)$, define the empirical loss

$$\ell_n(\theta) := -\frac{1}{n}\sum_{i=1}^{n} \log p(x_i \mid \theta),$$

where $p(x \mid \theta)$ is the model likelihood. The population loss is $\ell(\theta) := -\mathbb{E}_q[\log p(x \mid \theta)]$. Let θ^* denote a (possibly non-unique) minimizer of ℓ, and define the *excess loss*:

$$K(\theta) := \ell(\theta) - \ell(\theta^*) = D_{\mathrm{KL}}(q\|p_\theta) - D_{\mathrm{KL}}(q\|p_{\theta^*}).$$

This is the KL gap relative to the best approximation in the model. In the *realizable* case where $q = p_{\theta^*}$ for some θ^*, we have $K(\theta) = D_{\mathrm{KL}}(q\|p_\theta)$.

2.2 Local evidence and free energy

For a neighborhood $W \subset \Theta$ around θ^*, define the local evidence and free energy:

$$Z_n(W) := \int_W \exp\big(-n\ell_n(\theta)\big)\phi(\theta)\,d\theta, \tag{1}$$

$$F_n(W) := -\log Z_n(W). \tag{2}$$

2.3 Local learning coefficient (LLC)

The central object in SLT is the *real log canonical threshold* (RLCT), which we call the *local learning coefficient* (LLC).

Definition 1 (LLC via zeta function)**.** *For $K(\theta) \geq 0$ analytic near θ^* with $K(\theta^*) = 0$, define the zeta function*

$$\zeta(z) := \int_W K(\theta)^z \phi(\theta)\, d\theta$$

for $\mathrm{Re}(z) > 0$. This extends meromorphically to $\mathbb{C}$ with poles on the negative real axis. The LLC is $\lambda := -$ (largest pole), and the multiplicity *m is the order of this pole.*

Under standard regularity conditions (analyticity of K, positivity of ϕ near θ^*), SLT yields the asymptotic expansion:

$$F_n(W) = n\ell_n(\theta^*) + \lambda \log n - (m-1)\log\log n + O_p(1). \tag{3}$$

Equivalently, the sublevel-set volume satisfies

$$V(\varepsilon) := \mathrm{Vol}\{\theta \in W : K(\theta) \leq \varepsilon\} \sim c\,\varepsilon^{\lambda}(-\log\varepsilon)^{m-1} \quad \text{as } \varepsilon \to 0^+. \tag{4}$$

2.4 Standard additivity under independence

The following classical result motivates our development.

Proposition 1 (Additivity under independence)**.** *Suppose $\theta = (\theta^{(1)}, \ldots, \theta^{(K)})$ with each $\theta^{(k)} \in \Theta_k$, the prior factorizes as $\phi(\theta) = \prod_k \phi_k(\theta^{(k)})$, and the excess loss decomposes as $K(\theta) = \sum_k K_k(\theta^{(k)})$. Then $\lambda = \sum_k \lambda_k$, where λ_k is the LLC computed from K_k alone.*

Proof. We use sublevel-set volumes rather than direct zeta factorization (note: $(\sum_k K_k)^z \neq \prod_k K_k^z$, so the zeta function does not directly factor for sum losses).

The sublevel-set volume for the sum is

$$V(\varepsilon) = \mathrm{Vol}\{\theta : \textstyle\sum_k K_k(\theta^{(k)}) \leq \varepsilon\}.$$

With independent parameters and factorized prior, this becomes a convolution: if $V_k(\varepsilon) \sim c_k \varepsilon^{\lambda_k} |\log\varepsilon|^{m_k - 1}$ for each block, then

$$V(\varepsilon) = \int_{\sum_k \varepsilon_k = \varepsilon} \prod_k dV_k(\varepsilon_k) \sim c\,\varepsilon^{\sum_k \lambda_k} |\log\varepsilon|^{M-1},$$

where $M = \sum_k m_k + (K-1)$ accounts for the convolution. The Laplace-Stieltjes transform of V factors, and Tauberian theorems relate the transform's behavior to the volume scaling, yielding $\lambda = \sum_k \lambda_k$. □

3 SubRep Formalization

We formalize SubRep in loss-based language compatible with SLT.

3.1 Architecture and parameters

A SubRep system consists of the following parameterized components:

1. **Shared representation** $f_\psi : \mathcal{X} \to \mathcal{Z}$ with parameters $\psi \in \Psi$, mapping raw observations to features.
2. **Option models** $\{(\hat{r}_o, \hat{n}_o)\}_{o\in\mathcal{O}}$ with parameters $\theta_o \in \Theta_o$. Given features $z = f_\psi(x)$:
   - $\hat{r}_o(z; \theta_o)$: predicted immediate contribution to objective J;
   - $\hat{n}_o(z; \theta_o)$: predicted successor features.
3. **MDN** with parameters θ_{MDN}, producing motive geometry (cone W or finite cover W_{ref}).
4. **Planner** with parameters θ_{plan}, selecting options based on backed-up values.

The full parameter vector is

$$\Omega := (\psi, \{\theta_o\}_{o\in\mathcal{O}_{\text{lib}}}, \theta_{\text{MDN}}, \theta_{\text{plan}}).$$

3.2 Loss decomposition

The total loss $\ell(\Omega)$ naturally decomposes as:

$$\ell(\Omega) = \ell_{\text{rep}}(\psi) + \sum_{o\in\mathcal{O}_{\text{lib}}} \ell_o(\psi, \theta_o) + \ell_{\text{MDN}}(\theta_{\text{MDN}}) + \ell_{\text{plan}}(\psi, \{\theta_o\}, \theta_{\text{MDN}}, \theta_{\text{plan}}), \tag{5}$$

where:

- $\ell_{\text{rep}}(\psi)$: representation learning loss (e.g., reconstruction, contrastive);
- $\ell_o(\psi, \theta_o)$: option-o prediction loss (depends on shared features);
- ℓ_{MDN}: MDN auxiliary loss;
- ℓ_{plan}: planning loss (depends on all upstream components).

Remark 1 (Functional composition). The planner loss ℓ_{plan} depends on option outputs $\hat{r}_o, \hat{n}_o$, which in turn depend on ψ, θ_o. This creates *functional coupling*: gradients $\nabla_{\theta_o}\ell_{\text{plan}}$ flow through the composition. This coupling is the primary obstacle to naive LLC additivity.

3.3 CDS and PDS admission

Given weight vector w and features z, the backed-up value is

$$B_o(z; w) := \hat{r}_o(z) + w^\top \hat{n}_o(z). \tag{6}$$

Definition 2 (Cone-dominant subtask (CDS))**.** *Option o is CDS-admitted at margin $\varepsilon \geq 0$ if*

$$\inf_{w\in W} \big(B_o(z; w) - \max_{b\in B} B_b(z; w)\big) \geq \varepsilon$$

over a suitable distribution of states z.

Definition 3 (Pareto-dominant subtask (PDS))**.** *Given reference weights $W_{\text{ref}} = \{w^{(1)}, \ldots, w^{(K)}\}$, option o is PDS-admitted if the vector $(B_o(z; w^{(1)}), \ldots, B_o(z; w^{(K)}))$ Pareto-dominates the baseline vector.*

These tests involve inf, max, and inequality comparisons—all operations that introduce non-differentiability into the loss landscape.

4 Functional Coupling Analysis

We now analyze how functional coupling between planner and options affects LLC decomposition.

4.1 Coupling structure

Write the excess loss as

$$K(\Omega) = K_{\text{rep}}(\psi) + \sum_o K_o(\psi, \theta_o) + K_{\text{MDN}}(\theta_{\text{MDN}}) + K_{\text{plan}}(\Omega) + R(\Omega), \tag{7}$$

where $R(\Omega)$ captures cross-terms not assignable to individual modules.

The planner loss K_{plan} depends on option outputs. Expand around the optimum Ω^*:

$$K_{\text{plan}}(\Omega) = K^{(0)}_{\text{plan}}(\theta_{\text{plan}}) + \sum_o C_o(\theta_o, \theta_{\text{plan}}) + \text{h.o.t.},$$

where C_o captures the coupling between option o and the planner.

4.2 Coupling strength measure

Definition 4 (Coupling strength)**.** *The* coupling strength *of option o to the planner is*

$$\kappa_o := \limsup_{\Omega \to \Omega^*, \Omega \neq \Omega^*} \frac{|C_o(\theta_o, \theta_{\text{plan}})|}{K_o(\psi, \theta_o) + K^{(0)}_{\text{plan}}(\theta_{\text{plan}})},$$

the limiting ratio as parameters approach the optimum along paths where the denominator is nonzero. This captures the relative magnitude of coupling near the singular locus.

Definition 5 (Total coupling ratio)**.** *The* total coupling ratio *is*

$$\eta := \sup_{\Omega \in U} \frac{|R(\Omega)|}{K_0(\Omega)},$$

where $K_0(\Omega)$ is the sum of individual module excess losses.

4.3 Domination conditions

Assumption 1 (Dominated coupling)**.** *In a neighborhood U of Ω^*:*

(a) $\eta < 1$: the remainder is dominated by the sum of module losses.

(b) $\sum_o \kappa_o < 1 - \eta$: coupling strengths are collectively bounded.

(c) The prior $\phi(\Omega)$ factorizes up to bounded factors: $c_1 \prod_k \phi_k(\theta^{(k)}) \leq \phi(\Omega) \leq c_2 \prod_k \phi_k(\theta^{(k)})$ for constants $0 < c_1 \leq c_2 < \infty$.

Lemma 1 (Volume bound under dominated coupling)**.** *Under Assumption 1, the sublevel-set volume satisfies*

$$c_- \prod_k V_k(\varepsilon) \leq V(\varepsilon) \leq c_+ \prod_k V_k(\varepsilon)$$

for constants $c_-, c_+ > 0$ independent of ε, where $V_k(\varepsilon) := \mathrm{Vol}\{\theta^{(k)} : K_k(\theta^{(k)}) \leq \varepsilon\}$.

Proof. Under (a), $(1-\eta)K_0 \le K \le (1+\eta)K_0$. Thus $\{K \le \varepsilon\} \subseteq \{K_0 \le \varepsilon/(1-\eta)\}$ and $\{K_0 \le \varepsilon/(1+\eta)\} \subseteq \{K \le \varepsilon\}$.

For the upper bound:

$$\begin{aligned} V(\varepsilon) &\le \mathrm{Vol}\{K_0 \le \varepsilon/(1-\eta)\} \\ &= \mathrm{Vol}\Big\{\sum_k K_k \le \varepsilon/(1-\eta)\Big\} \\ &\le \prod_k V_k\big(\varepsilon/(1-\eta)\big) \end{aligned}$$

by inclusion-exclusion. The lower bound follows similarly. Since each $V_k(\varepsilon) \sim c_k \varepsilon^{\lambda_k} |\log \varepsilon|^{m_k - 1}$, the product scaling is preserved up to constants. □

4.4 Sufficient conditions on SubRep architecture

We now identify architectural conditions ensuring dominated coupling.

Proposition 2 (Conditions for dominated coupling in SubRep)**.** *The following conditions jointly imply Assumption 1:*

1. ***Option accuracy.*** *Each option model $(\hat{r}_o, \hat{n}_o)$ achieves loss $K_o \le \epsilon_{\mathrm{opt}}$ for small ϵ_{opt}.*
2. ***Planner smoothness.*** *The planner loss ℓ_{plan} is Lipschitz in option outputs with constant L_{plan}.*
3. ***Output sensitivity.*** *Option outputs are Lipschitz in parameters: $\|\hat{r}_o(\theta_o) - \hat{r}_o(\theta'_o)\| \le L_o \|\theta_o - \theta'_o\|$.*
4. ***Separation.*** *Options operate on distinct aspects of the task, limiting cross-option gradients: $\|\nabla_{\theta_o} \ell_{o'}\| \le \gamma \|\nabla_{\theta_o} \ell_o\|$ for $o \ne o'$ with $\gamma < 1/(|\mathcal{O}| - 1)$.*

Under these conditions, $\eta \le L_{\mathrm{plan}} \cdot L_{\max} \cdot \sqrt{\epsilon_{\mathrm{opt}}} + |\mathcal{O}|\gamma < 1$ for sufficiently small ϵ_{opt} and γ.

Proof sketch. The planner coupling term C_o arises from the composition of planner loss with option outputs. A key observation is that the *leading* linear coupling typically vanishes at the joint optimum (by first-order conditions), leaving C_o as a *second-order* cross-term.

Specifically, expand ℓ_{plan} around the optimum:

$$\ell_{\mathrm{plan}}(\hat{r}_o(\theta_o), \theta_{\mathrm{plan}}) = \ell^*_{\mathrm{plan}} + \underbrace{\nabla \ell \cdot \Delta \hat{r}_o}_{=0 \text{ at optimum}} + \frac{1}{2} \Delta \hat{r}_o^\top H \Delta \hat{r}_o + \cdots$$

The linear term vanishes by optimality. The quadratic term gives $|C_o| = O(\|\Delta \hat{r}_o\|^2) = O(L_o^2 \|\theta_o - \theta_o^*\|^2)$.

In *regular* directions where $K_o \sim \|\theta_o - \theta_o^*\|^2$, this yields $|C_o| = O(K_o)$, ensuring domination. In singular directions where K_o vanishes faster, the coupling ratio $|C_o|/K_o$ may grow, but Assumption 1 bounds this uniformly.

The separation condition bounds cross-module terms similarly. Combining these yields the stated bound on η. □

5 Shared Representations

Modern SubRep implementations typically share a feature extractor across options and planners. We analyze how this affects LLC decomposition.

5.1 Shared parameter structure

With shared representation f_ψ, the parameter space is *not* a product. Instead:

$$\Omega = (\psi, \theta_{\text{head}}) \quad \text{where} \quad \theta_{\text{head}} := (\{\theta_o^{\text{head}}\}_o, \theta_{\text{MDN}}, \theta_{\text{plan}}^{\text{head}}).$$

Here θ_o^{head} are option-specific heads operating on features $z = f_\psi(x)$.

5.2 Loss structure with sharing

The total loss becomes

$$\ell(\Omega) = \ell_{\text{rep}}(\psi) + \sum_o \ell_o^{\text{head}}(\theta_o^{\text{head}}; f_\psi) + \ell_{\text{plan}}^{\text{head}}(\theta_{\text{plan}}^{\text{head}}; f_\psi, \{\hat{r}_o, \hat{n}_o\}).$$

Crucially, ψ enters *every* term.

5.3 Intersection of singular structures

The singularity of $K(\Omega)$ at Ω^* is determined by the *intersection* of singularities from each loss component. This is more complex than the additive case.

Definition 6 (Singular variety)**.** *The* singular variety *of the excess loss is*

$$\mathcal{S} := \{\Omega : K(\Omega) = 0\} = K^{-1}(0).$$

With shared parameters, $\mathcal{S}$ is determined by the intersection:

$$\mathcal{S} = \bigcap_{\text{components } c} \{K_c = 0\},$$

where the intersection is taken in the full parameter space (not a product).

5.4 LLC for shared architectures

Theorem 1 (LLC with shared representations)**.** *Suppose the shared representation f_ψ satisfies:*

(a) ***Non-degenerate features.*** *The feature map f_ψ has full rank Jacobian at ψ^*.*

(b) ***Head separability.*** *Conditioned on features $z = f_\psi(x)$, the head losses ℓ_o^{head} are independent across options.*

Then the goal-level LLC satisfies

$$\lambda_{\text{goal}} = \lambda_\psi + \sum_o \lambda_o^{\text{head}} + \lambda_{\text{MDN}} + \lambda_{\text{plan}}^{\text{head}} + \lambda_{\text{int}}, \tag{8}$$

where λ_ψ is the LLC of the representation block, λ_o^{head} are the head LLCs computed with frozen features, and $\lambda_{\text{int}} \geq 0$ is an intersection correction *that vanishes when the singular varieties intersect transversally.*

Proof outline. Under (a), near ψ^* we can locally parameterize by $(z_\perp, z_\parallel)$ where $z_\perp$ spans directions in feature space and $z_\parallel$ are redundant parameters. The Jacobian condition ensures this reparameterization is valid.

Under (b), given fixed features, the head losses are conditionally independent. We use a stratified sublevel-set analysis: the total excess loss near the optimum can be written as $K = K_\psi + \sum_o K_o^{\text{head}} + K_{\text{MDN}} + K_{\text{plan}}^{\text{head}} + R_{\text{int}}$, where R_{int} captures cross-terms from non-transverse intersection of singular varieties.

When singular varieties intersect transversally (the generic case), R_{int} is higher-order dominated and contributes $\lambda_{\text{int}} = 0$. By the sublevel-set convolution argument of Proposition 1, the LLCs add. Non-transverse intersections (tangencies) make R_{int} contribute at leading order, increasing $\lambda_{\text{int}} > 0$. □

Remark 2. The key insight is that with sharing, the representation block's LLC λ_ψ contributes *once* to the total, not once per module. This differs from naive summation over modules.

6 Soft CDS/PDS and Stratified Analysis

CDS and PDS tests involve inf, max, and hard inequalities, introducing non-differentiable boundaries. Standard SLT requires analyticity. We address this via soft relaxations and stratified analysis.

6.1 Non-analyticity in hard CDS/PDS

The CDS margin function

$$m_o(z) := \inf_{w \in W} \big(B_o(z; w) - \max_{b \in B} B_b(z; w)\big)$$

involves inf over w and max over b. Both operations introduce kinks (non-differentiable points) in the loss landscape.

If CDS admission affects the loss (e.g., by gating which options receive gradient signal), then $K(\Omega)$ inherits these non-analyticities.

6.2 Soft relaxation

We introduce temperature-parameterized soft versions.

Definition 7 (Soft CDS). *For temperature $\tau > 0$, define the soft margin*

$$m_o^{(\tau)}(z) := -\tau \log \mathbb{E}_{w \sim \mu_W} \exp\big(-[B_o(z; w) - \mathrm{LSE}_b^{(\tau)} B_b(z; w)]/\tau\big),$$

where μ_W is a measure on W and $\mathrm{LSE}^{(\tau)}$ is the log-sum-exp (soft max): $\mathrm{LSE}_b^{(\tau)} a_b := \tau \log \sum_b \exp(a_b/\tau)$.

Lemma 2 (Soft-to-hard convergence). *As $\tau \to 0^+$:*

1. $\mathrm{LSE}_b^{(\tau)} a_b \to \max_b a_b$ *pointwise.*
2. $m_o^{(\tau)}(z) \to m_o(z)$ *pointwise.*
3. *If $K^{(\tau)}$ denotes the excess loss with soft CDS/PDS, then $K^{(\tau)} \to K$ pointwise.*

Proof. Standard properties of log-sum-exp approximations. □

6.3 LLC for soft systems

For $\tau > 0$, the soft loss $K^{(\tau)}(\Omega)$ is analytic (assuming the underlying models are analytic). Thus SLT applies, yielding a well-defined LLC $\lambda^{(\tau)}$.

Definition 8 (Stratified LLC)**.** *The* stratified LLC *of the hard system is*

$$\lambda := \lim_{\tau \to 0^+} \lambda^{(\tau)},$$

provided the limit exists.

6.4 Stratification structure

The hard CDS/PDS conditions partition parameter space into strata based on the active set of admitted options and active constraints.

Definition 9 (CDS/PDS stratification)**.** *Define the* active set *function* $A(\Omega) := \{o \in \mathcal{O} : o$ *is CDS/PDS admitted*
The parameter space stratifies as

$$\Theta = \bigsqcup_{S \subseteq \mathcal{O}} \Sigma_S, \quad \Sigma_S := \{\Omega : A(\Omega) = S\}.$$

Theorem 2 (Existence of stratified LLC)**.** *Assume:*

(a) On each stratum Σ_S, the loss $K|_{\Sigma_S}$ is analytic.

(b) Strata boundaries are semi-algebraic (defined by polynomial inequalities).

(c) The prior ϕ is positive and analytic.

Then:

1. *The limit $\lambda = \lim_{\tau \to 0^+} \lambda^{(\tau)}$ exists.*
2. *λ equals the LLC computed on the stratum Σ_{S^*} containing Ω^*, provided Ω^* is in the interior of Σ_{S^*}.*
3. *If Ω^* lies on a stratum boundary, λ is determined by the resolution of singularities accounting for the boundary.*

Proof sketch. Under (b), the stratification is a Whitney stratification of a semi-algebraic set. Resolution of singularities (Hironaka) provides a principalization where the loss becomes monomial-like in local coordinates. The zeta function method extends to this setting, yielding a well-defined LLC.

The soft approximation $K^{(\tau)}$ smooths across strata. As $\tau \to 0$, the dominant contribution to the zeta integral concentrates on the stratum containing Ω^*. Standard arguments (Laplace method) show $\lambda^{(\tau)} \to \lambda$. ∎

6.5 LLC jumps at stratum boundaries

Proposition 3 (Discontinuity structure)**.** *Consider a path $\Omega(t)$ crossing from stratum Σ_{S_1} to Σ_{S_2} at $t = t_0$. If the singular varieties on the two strata have different dimensions or intersection structures, then the LLC function $t \mapsto \lambda(\Omega(t))$ is discontinuous at t_0.*

Proof. The LLC is determined by the dimension and intersection multiplicity of the singular variety. At a stratum boundary, the loss function changes (different options are active), potentially altering these geometric quantities. The LLC inherits this discontinuity. □

Example 1 (Option admission boundary). Suppose at Ω^*, option o is exactly at the CDS admission threshold: $m_o(z) = 0$. For Ω slightly perturbed:

- If $m_o > 0$: option o is admitted, contributing λ_o to the total.
- If $m_o < 0$: option o is excluded, and its λ_o does not contribute.

Thus λ_{goal} jumps by $\pm\lambda_o$ across the boundary.

7 Non-Convex Motive Geometries

The preceding analysis assumed convex motive cones W. However, motive geometries learned by MDNs are typically non-convex. This section develops SLT for general motive geometries, revealing rich structure connecting to tropical geometry and polyhedral combinatorics.

7.1 Motivation: Why non-convexity matters

In practice, motive geometries arise from:

1. **Learned MDN outputs.** Neural networks produce non-convex sublevel sets and decision boundaries.
2. **Compositional motives.** Combining multiple motive factors via nonlinear operations (products, mins, maxes) destroys convexity.
3. **Constraint satisfaction.** Hard constraints on acceptable trade-offs create non-convex feasible regions.
4. **Multi-modal preferences.** Agents with distinct "preference modes" have disconnected or non-convex motive regions.

Non-convexity fundamentally changes the CDS margin function $m_o(z) = \inf_{w\in W}(B_o(z;w) - B_{\text{base}}(z;w))$: the infimum over a non-convex set can have multiple local minima, creating additional non-smoothness beyond the max over baselines.

7.2 Convex decomposition of motive geometries

We begin by decomposing a general motive geometry into convex pieces.

Definition 10 (Convex decomposition)**.** *A* convex decomposition *of a compact set $W \subset \mathbb{R}^d$ is a finite collection $\{W_\alpha\}_{\alpha\in\mathcal{A}}$ of closed convex sets such that:*

1. *$W = \bigcup_\alpha W_\alpha$ (coverage);*
2. *$\operatorname{int}(W_\alpha) \cap \operatorname{int}(W_\beta) = \emptyset$ for $\alpha \neq \beta$ (interior disjointness);*
3. *Each W_α is a convex polytope (or more generally, a convex body).*

The cell complex *$\mathcal{C}(W)$ consists of the cells $\{W_\alpha\}$ together with their faces.*

Definition 11 (Boundary complex)**.** *The* boundary complex $\partial\mathcal{C}(W)$ *consists of all faces of dimension* $< d$ *in* $\mathcal{C}(W)$*. A point* $w \in W$ *is a* boundary point *if* $w \in |\partial\mathcal{C}(W)|$*, where* $|\cdot|$ *denotes the underlying point set.*

Remark 3 (Existence of convex decompositions). For semi-algebraic W (definable by polynomial inequalities), a convex decomposition always exists by triangulation. For general compact W, approximate convex decompositions exist to arbitrary precision. In practice, MDN-induced geometries are piecewise-defined and admit natural decompositions.

7.3 Per-cell CDS and the piecewise margin function

With a convex decomposition $\{W_\alpha\}$, define per-cell margins:

Definition 12 (Cell-restricted margin)**.** *For cell* W_α*, the* cell-restricted margin *is*

$$m_o^\alpha(z) := \inf_{w \in W_\alpha} \big(B_o(z; w) - B_{\text{base}}(z; w)\big).$$

Since each W_α is convex, Proposition 2 and the soft CDS analysis apply per-cell.

Proposition 4 (Piecewise structure of global margin)**.** *The global margin function satisfies*

$$m_o(z) = \min_{\alpha \in \mathcal{A}} m_o^\alpha(z). \tag{9}$$

This is a piecewise function: for each z*, there exists an* active cell $\alpha^*(z) := \arg\min_\alpha m_o^\alpha(z)$ *(possibly non-unique) achieving the minimum.*

Proof. By definition, $m_o(z) = \inf_{w \in W}(\cdots)$. Since $W = \bigcup_\alpha W_\alpha$:

$$\inf_{w \in W} f(w) = \inf_\alpha \inf_{w \in W_\alpha} f(w) = \min_\alpha m_o^\alpha(z).$$

□

7.4 Tropical structure and min-plus algebra

The piecewise-min structure (9) connects to *tropical geometry*, the geometry of piecewise-linear functions under $(\min, +)$ operations.

Definition 13 (Tropical margin)**.** *View the margin as a function* $m_o : \mathcal{Z} \to \mathbb{R} \cup \{+\infty\}$ *in the* tropical semiring $(\mathbb{R} \cup \{+\infty\}, \min, +)$*. The global margin is the tropical sum (pointwise minimum):*

$$m_o = \bigoplus_\alpha m_o^\alpha := \min_\alpha m_o^\alpha.$$

Definition 14 (Tropical hypersurface)**.** *The* tropical hypersurface $\mathcal{T}_o \subset \mathcal{Z}$ *of option* o *is the set of points where the minimum in* (9) *is achieved by multiple cells:*

$$\mathcal{T}_o := \{z \in \mathcal{Z} : |\arg\min_\alpha m_o^\alpha(z)| \geq 2\}.$$

The tropical hypersurface is where the margin function is non-smooth due to cell transitions.

Proposition 5 (Structure of tropical hypersurface)**.** *If each* $m_o^\alpha(z)$ *is piecewise-affine in* z*, then* $\mathcal{T}_o$ *is a polyhedral complex of codimension* ≥ 1*.*

Proof. Each m_o^α is piecewise-affine, hence $\min_\alpha m_o^\alpha$ is piecewise-affine. The non-smooth locus of a piecewise-affine function is a polyhedral complex where pieces meet. By genericity (distinct affine pieces), this has codimension ≥ 1. □

7.5 Double stratification: options $\times$ cells

Non-convex motive geometry introduces a *double stratification*: the parameter space stratifies by both (i) which options are admitted and (ii) which motive cells are active.

Definition 15 (Active cell function)**.** *For parameter Ω and state distribution, define the* active cell set

$$C(\Omega) := \{\alpha \in \mathcal{A} : \alpha = \alpha^*(z) \text{ for } z \text{ in the support}\}.$$

This is the set of cells that are "active" (achieve the margin minimum) somewhere in the state distribution.

Definition 16 (Double stratification)**.** *The* double stratification *of parameter space is*

$$\Theta = \bigsqcup_{(S,\mathcal{B})} \Sigma_{S,\mathcal{B}},$$

where:

- *$S \subseteq \mathcal{O}$ is the set of admitted options;*
- *$\mathcal{B} \subseteq \mathcal{A}$ is the set of active cells;*
- *$\Sigma_{S,\mathcal{B}} := \{\Omega : A(\Omega) = S,\, C(\Omega) = \mathcal{B}\}$.*

Remark 4. The double stratification is finer than the option-only stratification of Section 6. The number of strata is bounded by $2^{|\mathcal{O}|} \times 2^{|\mathcal{A}|}$, though many combinations may be empty.

7.6 LLC on double strata

On each double stratum $\Sigma_{S,\mathcal{B}}$, the effective loss has a specific structure determined by $(S, \mathcal{B})$.

Definition 17 (Stratum-restricted loss)**.** *On stratum $\Sigma_{S,\mathcal{B}}$, define the effective excess loss*

$$K_{S,\mathcal{B}}(\Omega) := K_{\text{rep}}(\psi) + \sum_{o \in S} K_o(\psi, \theta_o) + K_{\text{MDN}}(\theta_{\text{MDN}}) + K_{\text{plan}}^{S,\mathcal{B}}(\Omega),$$

where $K_{\text{plan}}^{S,\mathcal{B}}$ is the planning loss restricted to admitted options S and active cells $\mathcal{B}$.

Theorem 3 (LLC on double strata)**.** *Assume:*

(a) On each double stratum $\Sigma_{S,\mathcal{B}}$, the loss $K_{S,\mathcal{B}}$ is analytic.

(b) Double stratum boundaries are semi-algebraic.

(c) Dominated coupling holds within each stratum.

Then the LLC on stratum $\Sigma_{S,\mathcal{B}}$ satisfies

$$\lambda_{S,\mathcal{B}} = \lambda_\psi + \sum_{o \in S} \lambda_o^{\text{head}} + \lambda_{\text{MDN}} + \lambda_{\text{plan}}^{S,\mathcal{B}} + \lambda_{\text{int}}^{S,\mathcal{B}}, \tag{10}$$

where $\lambda_{\text{plan}}^{S,\mathcal{B}}$ depends on the specific combination of admitted options and active cells, and $\lambda_{\text{int}}^{S,\mathcal{B}}$ is the intersection correction.

Proof. Apply Theorem 2 to the double stratification. The semi-algebraic condition ensures the stratification is Whitney. Per-stratum analyticity and dominated coupling give the decomposition by Theorem 1. $\square$

7.7 Cell boundary singularities

Transitions between active cells create additional singularities beyond option admission boundaries.

Definition 18 (Cell transition boundary)**.** *A* cell transition boundary *is a codimension-1 subset of parameter space where the active cell set changes:*

$$\partial_{\text{cell}} := \{\Omega : C(\Omega) \text{ changes under infinitesimal perturbation}\}.$$

Proposition 6 (Singularity at cell boundaries)**.** *At a cell transition boundary where the active set changes from $\mathcal{B}_1$ to $\mathcal{B}_2 = \mathcal{B}_1 \cup \{\alpha'\}$ (a single cell α' becomes active):*

1. *The margin function $m_o(z)$ has a kink (non-differentiable point) at the boundary.*
2. *If the planning loss depends on margin gradients, the loss $K(\Omega)$ inherits this non-smoothness.*
3. *The LLC may jump: $\lambda_{S,\mathcal{B}_2} - \lambda_{S,\mathcal{B}_1} = \Delta\lambda_{\alpha'}$, where $\Delta\lambda_{\alpha'}$ is the LLC contribution from cell α'.*

Proof. (1) At the boundary, $m_o^\alpha(z) = m_o^{\alpha'}(z)$ for some $\alpha \in \mathcal{B}_1$. The function $\min(m_o^\alpha, m_o^{\alpha'})$ has a kink where the two terms are equal.

(2) If K_{plan} involves $\nabla_\theta m_o$, the non-differentiability propagates.

(3) The singular variety geometry changes when a new cell becomes active, potentially altering the LLC. The jump magnitude depends on how cell α' contributes to the singularity structure. □

7.8 Soft relaxation for non-convex geometries

We extend the soft CDS/PDS construction to non-convex geometries.

Definition 19 (Soft non-convex margin)**.** *For temperature $\tau > 0$, define the soft margin over non-convex W:*

$$m_o^{(\tau)}(z) := -\tau \log \sum_{\alpha \in \mathcal{A}} \exp\big(-m_o^\alpha(z)/\tau\big).$$

This is a soft-min (log-sum-exp) over cells.

Proposition 7 (Soft-to-hard convergence for non-convex)**.** *As $\tau \to 0^+$:*

1. *$m_o^{(\tau)}(z) \to \min_\alpha m_o^\alpha(z) = m_o(z)$ pointwise.*
2. *$m_o^{(\tau)}$ is smooth for $\tau > 0$.*
3. *The gradient $\nabla_\theta m_o^{(\tau)}$ is a convex combination of per-cell gradients:*

   $$\nabla_\theta m_o^{(\tau)} = \sum_\alpha p_\alpha^{(\tau)}(z) \nabla_\theta m_o^\alpha,$$

   where $p_\alpha^{(\tau)}(z) \propto \exp(-m_o^\alpha(z)/\tau)$ are softmax weights.

Proof. Standard properties of log-sum-exp. The gradient formula follows from differentiating $-\tau \log \sum_\alpha \exp(-m_o^\alpha/\tau)$. □

Corollary 1 (Soft double-stratum LLC)**.** *For $\tau > 0$, the soft loss $K^{(\tau)}$ is analytic, with LLC $\lambda^{(\tau)}$. As $\tau \to 0$:*

$$\lambda^{(\tau)} \to \lambda_{S^*,\mathcal{B}^*},$$

the LLC of the double stratum containing the optimum Ω^*.

7.9 Tropical LLC and Bergman fans

For piecewise-linear margins, the LLC structure connects to *tropical intersection theory.*

Definition 20 (Tropical LLC contribution)**.** *Consider the tropical hypersurface $\mathcal{T} = \bigcup_o \mathcal{T}_o$ (union over options). The* tropical LLC contribution *is*

$$\lambda_{\mathrm{trop}} := \sum_{\sigma \in \mathcal{T}} \mu(\sigma) \cdot \lambda_\sigma,$$

where the sum is over cells σ of $\mathcal{T}$, $\mu(\sigma)$ is the multiplicity (number of affine pieces meeting at σ), and λ_σ is the LLC contribution from singularities along σ.

Theorem 4 (Tropical correction to LLC)**.** *Under the conditions of Theorem 3, the goal-level LLC satisfies*

$$\lambda_{\mathrm{goal}} = \lambda_{\mathrm{base}} + \lambda_{\mathrm{trop}}, \tag{11}$$

where λ_{base} is the LLC computed ignoring tropical singularities (as if the geometry were convex), and $\lambda_{\mathrm{trop}} \geq 0$ is the tropical correction from cell boundary singularities.

Proof sketch. The singular variety of $K(\Omega)$ is stratified by the tropical hypersurface. On the complement of $\mathcal{T}$, the loss is smooth and contributes λ_{base}. Along each stratum of $\mathcal{T}$, additional singularities from the piecewise structure contribute λ_{trop}.

By resolution of singularities, the total LLC is the sum of contributions from all strata of the exceptional divisor. The tropical structure provides a combinatorial indexing of these strata. □

Remark 5 (Connection to Bergman fans). For matroids and linear spaces, the tropical variety is the *Bergman fan*, whose structure is well-understood combinatorially. When the motive geometry arises from matroid-like structures (e.g., submodular motives), the tropical LLC contribution can be computed from matroid invariants.

7.10 Bounds on tropical correction

Proposition 8 (Upper bound on λ_{trop})**.** *Assume:*

(i) The convex decomposition $\{W_\alpha\}$ has a tree-like *adjacency structure (i.e., the dual graph is a tree, so there are no cycles of adjacent cells).*

(ii) Cell boundaries intersect transversally *(generically).*

Then the tropical correction satisfies

$$\lambda_{\mathrm{trop}} \leq (|\mathcal{A}| - 1) \cdot \frac{d_W - 1}{2},$$

where $|\mathcal{A}|$ is the number of convex cells and d_W is the dimension of the weight space. Equality holds only in highly degenerate cases where the optimum lies on multiple cell boundaries simultaneously.

Proof sketch. Under (ii), each cell boundary (a codimension-1 interface) contributes at most $\frac{d_W-1}{2}$ to the LLC—the RLCT of a generic fold singularity in d_W dimensions.

Under (i), a tree on $|\mathcal{A}|$ vertices has exactly $|\mathcal{A}| - 1$ edges, hence at most $|\mathcal{A}| - 1$ cell boundaries. Summing contributions gives the bound.

If assumption (i) fails (the cell complex has cycles), additional boundaries exist and the bound increases. If assumption (ii) fails (tangent boundaries), individual contributions can exceed $\frac{d_W-1}{2}$. □

Corollary 2 (Negligibility condition)**.** *The tropical correction is negligible ($\lambda_{\text{trop}} \ll \lambda_{\text{base}}$) when:*

1. *The number of cells $|\mathcal{A}|$ is small.*
2. *Cell boundaries are transverse (non-degenerate intersections).*
3. *The optimum Ω^* is far from cell boundaries.*

Under these conditions, the convex analysis is a good approximation.

7.11 Example: Two-cell motive geometry

Example 2 (Binary motive structure). Consider $W = W_1 \cup W_2$ with two convex cells meeting along a hyperplane $H = W_1 \cap W_2$. Suppose an option o has:

- Cell-1 margin: $m_o^1(z) = a_1^\top z + b_1$;
- Cell-2 margin: $m_o^2(z) = a_2^\top z + b_2$.

The global margin is $m_o(z) = \min(m_o^1(z), m_o^2(z))$, with tropical hypersurface $\mathcal{T}_o = \{z : a_1^\top z + b_1 = a_2^\top z + b_2\}$.

LLC structure:

- Away from $\mathcal{T}_o$: LLC decomposes per the active cell.
- On $\mathcal{T}_o$: Additional singularity from the kink. If $a_1 \neq a_2$, this is a generic fold with LLC contribution $\lambda_{\text{fold}} = \frac{1}{2}$.

The total LLC decomposes as:

$$\lambda_{\text{goal}} = \lambda_{\text{base}} + \frac{1}{2} \cdot \mathbf{1}[\Omega^* \text{ on } \mathcal{T}_o],$$

where $\mathbf{1}[\cdot]$ is an indicator. When Ω^* is strictly off the tropical hypersurface, the LLC is *exactly* λ_{base}, not approximately. The jump of 1/2 occurs precisely when Ω^* lies on $\mathcal{T}_o$.

7.12 Computational considerations

Remark 6 (Practical computation for non-convex geometries). Computing the LLC for non-convex motive geometries requires:

1. **Cell decomposition.** Compute or approximate a convex decomposition of W. For neural MDNs, this can be done by linearizing around operating points.
2. **Per-cell LLC.** Compute λ_α for each cell using the convex analysis.
3. **Tropical structure.** Identify the tropical hypersurface and compute its contribution.
4. **Active cell determination.** For given Ω, determine which cells are active to identify the relevant stratum.

In practice, if the number of cells is moderate and the optimum is not near cell boundaries, the convex approximation ($\lambda \approx \lambda_{\text{base}}$) is often adequate.

8 Main Decomposition Theorems

We now state our main results combining the preceding analyses, including non-convex motive geometries.

Theorem 5 (LLC decomposition for SubRep: General case)**.** *Consider a SubRep system with shared representation and non-convex motive geometry* $W = \bigcup_\alpha W_\alpha$*, parameterized by* $\Omega = (\psi, \{\theta_o^{\text{head}}\}_o, \theta_{\text{MDN}}, \theta_{\text{plan}}^{\text{head}}$ *Assume:*

1. *Dominated coupling (Assumption 1).*
2. *Non-degenerate shared features and head separability (Theorem 1).*
3. *Per-double-stratum analyticity and semi-algebraic boundaries (Theorem 3).*
4. *The optimum* Ω^* *lies in the interior of a double stratum* $\Sigma_{S^*,\mathcal{B}^*}$*.*

Then the goal-level LLC satisfies

$$\lambda_{\text{goal}} = \lambda_\psi + \sum_{o \in S^*} \lambda_o^{\text{head}} + \lambda_{\text{MDN}} + \lambda_{\text{plan}}^{S^*,\mathcal{B}^*} + \lambda_{\text{int}} + \lambda_{\text{trop}}, \tag{12}$$

where:

- λ_ψ *is the LLC of the shared representation block;*
- $S^* = A(\Omega^*)$ *is the active set of admitted options;*
- $\mathcal{B}^* = C(\Omega^*)$ *is the active set of motive cells;*
- λ_o^{head} *is the LLC of option o's head (computed with frozen features);*
- $\lambda_{\text{plan}}^{S^*,\mathcal{B}^*}$ *is the planner LLC on the active stratum;*
- $\lambda_{\text{int}} \geq 0$ *is the intersection correction (zero under transverse intersection);*
- $\lambda_{\text{trop}} \geq 0$ *is the tropical correction from non-convex cell boundaries (zero if* Ω^* *is away from cell transitions).*

Proof. Combine Lemma 1, Theorem 1, Theorem 3, and Theorem 4. The dominated coupling ensures the remainder R does not affect the LLC order. The shared representation analysis accounts for parameter sharing. The double stratification analysis handles both option admission and motive cell transitions. The tropical correction captures singularities from piecewise motive geometry. □

Corollary 3 (Convex case recovery)**.** *If the motive geometry* W *is convex (a single cell,* $|\mathcal{A}| = 1$*), then* $\mathcal{B}^* = \{W\}$*,* $\lambda_{\text{trop}} = 0$*, and Theorem 5 reduces to:*

$$\lambda_{\text{goal}} = \lambda_\psi + \sum_{o \in S^*} \lambda_o^{\text{head}} + \lambda_{\text{MDN}} + \lambda_{\text{plan}} + \lambda_{\text{int}}.$$

Corollary 4 (Per-weight LLC vectors with non-convex geometry)**.** *For PDS with reference weights* $W_{\text{ref}} = \{w^{(1)}, \ldots, w^{(K)}\}$ *drawn from a non-convex motive geometry, define per-weight losses* $K^{(k)}$ *by restricting to weight* $w^{(k)}$*. Under the conditions of Theorem 5 applied per-weight:*

$$\Lambda_{\text{goal}} := (\lambda_{\text{goal}}^{(1)}, \ldots, \lambda_{\text{goal}}^{(K)})$$

with each component decomposing as in (12)*. The tropical correction* $\lambda_{\text{trop}}^{(k)}$ *depends on which cell contains* $w^{(k)}$ *and its proximity to cell boundaries.*

Theorem 6 (Robust decomposition bounds)**.** *Even when dominated coupling fails, the goal-level LLC satisfies*

$$\max_k \lambda_k \le \lambda_{\text{goal}} \le \sum_k \lambda_k + \lambda_R,$$

where λ_R is the LLC of the remainder term $R(\Omega)$ when it creates independent singularities. The upper bound holds when $R \ge 0$ and the losses are "separable enough" (see proof).

Proof. **Lower bound:** For any k, the total loss satisfies $K(\Omega) \ge K_k(\theta^{(k)})$ when all $K_j \ge 0$ and $R \ge 0$. Thus $\{K \le \varepsilon\} \subseteq \{K_k \le \varepsilon\}$, giving $V(\varepsilon) \le V_k(\varepsilon)$. Since $V_k(\varepsilon) \sim c_k \varepsilon^{\lambda_k}$, we have $\lambda \ge \lambda_k$ for each k.

Upper bound: This requires care. When R creates singularities independent of the K_k, we can decompose the singular variety: $\mathcal{S} = \{K = 0\} \subseteq \{K_0 = 0\} \cup \{R = 0\}$ where $K_0 = \sum_k K_k$. By the additivity of LLC for independent losses (Proposition 1), $\lambda_{K_0} = \sum_k \lambda_k$.

If R has singular variety $\mathcal{S}_R$ disjoint from $\mathcal{S}_0$, the total singularity is the union, and $\lambda_{\text{goal}} \le \lambda_{K_0} + \lambda_R$ by subadditivity of LLC over unions of singular varieties.

If $\mathcal{S}_R \cap \mathcal{S}_0 \ne \emptyset$, the intersection can reduce the total (redundancy) or increase it (if intersection creates higher-order tangencies). The bound λ_R is then an upper bound on the additional contribution, not necessarily tight.

Caveat: When R shares parameters with the K_k and creates correlated singularities, the upper bound may not be tight. In such cases, the interaction framework of Section 10 provides the correct decomposition. □

9 Robustness and Degradation Analysis

The preceding sections establish LLC additivity under "nice" conditions. This section addresses the critical practical question: *how quickly and severely does additivity break down as conditions degrade?* We characterize both gradual degradation (continuous deviation from additivity) and catastrophic failure (sharp phase transitions).

9.1 The additivity gap

Definition 21 (Additivity gap)**.** *The* additivity gap *is the deviation of the goal-level LLC from the sum of module LLCs:*

$$\Delta\lambda := \lambda_{\text{goal}} - \sum_k \lambda_k, \tag{13}$$

where the sum is over all modules (shared representation, option heads, MDN, planner). Under exact additivity, $\Delta\lambda = 0$.

From Theorem 5, under ideal conditions:

$$\Delta\lambda = \lambda_{\text{int}} + \lambda_{\text{trop}} \ge 0,$$

with both corrections small (often zero) under transversality and convexity assumptions. When conditions fail, $\Delta\lambda$ can grow substantially.

9.2 Coupling-induced deviation

We analyze how coupling affects the LLC. A key distinction is between *multiplicative perturbations* (which preserve LLC) and *structural perturbations* (which can change LLC).

Theorem 7 (LLC invariance under dominated coupling)**.** *Suppose the coupling ratio η satisfies $0 \leq \eta < 1$ and the remainder $R(\Omega)$ is* structurally subordinate *to K_0:*

(i) R vanishes on $\mathcal{S}_0 := \{K_0 = 0\}$, and

(ii) R/K_0 extends continuously to $\mathcal{S}_0$.

Then the asymptotic LLC satisfies $\lambda = \sum_k \lambda_k$ exactly. The coupling affects only the multiplicative constant, not the exponent.

Proof. From dominated coupling, $(1-\eta)K_0 \leq K \leq (1+\eta)K_0$. For sublevel-set volumes:

$$V_0(\varepsilon/(1+\eta)) \leq V(\varepsilon) \leq V_0(\varepsilon/(1-\eta)).$$

Since $V_0(\varepsilon) \sim c\varepsilon^{\sum_k \lambda_k} |\log \varepsilon|^{m-1}$, scaling ε by a constant factor changes the prefactor but not the exponent. Thus $V(\varepsilon) \sim c'\varepsilon^{\sum_k \lambda_k}$ with $c' \in [c(1+\eta)^{-\sum_k \lambda_k}, c(1-\eta)^{-\sum_k \lambda_k}]$. □

Remark 7 (When does LLC actually change?). The LLC changes only when the remainder R modifies the *singular structure*—specifically, when R has a different vanishing order than K_0 in some direction, or when R creates new singularities not present in K_0. This corresponds to $I_\lambda \neq 0$ in the interaction framework of Section 10.

Theorem 8 (Additivity gap characterization)**.** *The additivity gap $\Delta\lambda = \lambda_{\text{goal}} - \sum_k \lambda_k$ satisfies:*

1. *$\Delta\lambda = 0$ if the remainder R is structurally subordinate.*
2. *$|\Delta\lambda| \leq \lambda_R$ where λ_R is the LLC contribution from R treated as an independent loss (if R creates new singularities).*
3. *$\Delta\lambda < 0$ is possible if modules share singular structure (redundancy, $I_\lambda > 0$).*
4. *$\Delta\lambda > 0$ is possible if coupling creates new singularities (synergy, $I_\lambda < 0$).*

Remark 8 (Finite-sample effective LLC). In practice, with finite samples n, the *effective LLC* $\hat{\lambda}_n$ estimated from evidence or WBIC can vary with coupling even when the asymptotic LLC is unchanged. Finite-sample corrections scale as $O(1/\log n)$ and can be affected by the coupling-modified prefactor. The "linear degradation" intuition applies to these finite-sample estimates, not to the true asymptotic LLC.

9.3 Phase transition at the dominated boundary

When the remainder R transitions from structurally subordinate to structurally significant, the LLC can change discontinuously.

Theorem 9 (Phase transition in LLC)**.** *Consider a family of losses $K_\eta(\Omega) = K_0(\Omega) + \eta R(\Omega)$ parameterized by coupling strength $\eta \geq 0$. Define $\lambda(\eta) :=$ LLC of K_η. Then:*

1. *For $\eta < 1$ with R structurally subordinate to K_0: $\lambda(\eta) = \sum_k \lambda_k$ exactly (Theorem 7).*
2. *At $\eta = 1$: If R has singularities not present in K_0, then $\lambda(\eta)$ can jump discontinuously.*

3. *For $\eta > 1$: The LLC is determined by the combined singular geometry of K_0 and R, potentially differing from both $\sum_k \lambda_k$ and λ_R.*

Proof sketch. (1) Follows from Theorem 7.

(2) At $\eta = 1$, the remainder R contributes at the same order as K_0. If R vanishes on a different variety than K_0, the intersection of these varieties determines the new singular locus. This intersection can have different dimension, changing the LLC discontinuously.

(3) For $\eta > 1$, the remainder dominates: $K_\eta \approx \eta R$ for large η. The LLC approaches λ_R, the LLC of R alone. □

Example 3 (Sharp transition). Let $K_0(\theta) = \theta_1^2 + \theta_2^2$ (smooth, $\lambda_0 = 1$) and $R(\theta) = (\theta_1 - \theta_2)^2$ (singular along $\theta_1 = \theta_2$, $\lambda_R = 1/2$).

For $K_\eta = K_0 + \eta R = \theta_1^2 + \theta_2^2 + \eta(\theta_1 - \theta_2)^2$:

- $\eta < 1$: $\lambda(\eta) = 1$ (the K_0 structure dominates).
- $\eta = 1$: $K_1 = 2\theta_1^2 + 2\theta_2^2 - 2\theta_1\theta_2$, still smooth, $\lambda(1) = 1$.
- $\eta \gg 1$: $K_\eta \approx \eta(\theta_1 - \theta_2)^2$, so $\lambda(\eta) \to 1/2$.

In this case, the transition is gradual. But if R had a more complex singularity (e.g., $R = \theta_1^2\theta_2^2$), sharp jumps can occur.

9.4 Intersection correction growth

The intersection correction λ_{int} from shared representations can also grow as conditions degrade.

Proposition 9 (Intersection correction bounds)**.** *Let $d_\psi = \dim(\psi)$ be the shared representation dimension. The intersection correction satisfies:*

1. ***Transverse case:*** $\lambda_{\text{int}} = 0$.
2. ***Tangent intersection:*** $\lambda_{\text{int}} \le d_\psi/2$.
3. ***Worst case (full degeneracy):*** $\lambda_{\text{int}} \le \min(d_\psi, \sum_o d_o)$*, where* $d_o = \dim(\theta_o^{\text{head}})$.

Proof. (1) Transverse intersections contribute no additional singularity.

(2) A tangent intersection of two hypersurfaces in d_ψ dimensions has LLC at most $d_\psi/2$ (the LLC of a fold singularity).

(3) In the worst case, all singular varieties coincide on a subspace, giving LLC bounded by the smaller of the two dimensions involved. □

Remark 9 (Jacobian rank and intersection)*.* When the shared representation Jacobian $\partial f_\psi/\partial\psi$ drops rank, the intersection becomes non-transverse. If rank drops by r, expect $\lambda_{\text{int}} \approx r/2$.

9.5 Tropical correction growth

The tropical correction λ_{trop} from non-convex motive geometry also exhibits characteristic growth patterns.

Proposition 10 (Tropical correction scaling)**.** *Let $|\mathcal{A}|$ be the number of convex cells in the motive geometry and $d_W = \dim(W)$ the weight space dimension.*

1. ***Few cells, transverse boundaries:*** $\lambda_{\text{trop}} = O(|\mathcal{A}|)$ *with small constant.*

2. ***Many cells:*** *$\lambda_{\text{trop}} = O(|\mathcal{A}| \cdot d_W)$ in the worst case.*

3. ***Degenerate boundaries (tangencies):*** *Each k-fold tangency adds $O(k \cdot d_W)$ to λ_{trop}.*

Proof. (1) From Proposition 8, transverse boundaries contribute $\leq (d_W - 1)/2$ each, with $|\mathcal{A}| - 1$ boundaries.

(2) Non-transverse boundaries can contribute up to d_W each.

(3) A k-fold tangency (where k cells meet tangentially) amplifies the contribution by factor k. □

Corollary 5 (Cell count threshold). *For $\lambda_{\text{trop}} \lesssim 0.1\lambda_{\text{base}}$ (tropical correction under 10% of base), require*

$$|\mathcal{A}| \lesssim \frac{0.2\lambda_{\text{base}}}{d_W - 1}.$$

For a system with $\lambda_{\text{base}} = 10$ and $d_W = 5$, this gives $|\mathcal{A}| \lesssim 0.5$, i.e., the geometry must be essentially convex for small tropical corrections.

9.6 Stratum boundary effects

Proximity to stratum boundaries (option admission or cell transitions) creates additional deviation from smooth additivity.

Definition 22 (Boundary distance). *For parameter Ω in stratum $\Sigma_{S,\mathcal{B}}$, define the* boundary distance

$$\delta(\Omega) := \inf_{\Omega' \in \partial\Sigma_{S,\mathcal{B}}} \|\Omega - \Omega'\|,$$

the distance to the nearest stratum boundary.

Proposition 11 (Boundary proximity effects). *Assume the loss is* locally regular *near the stratum boundary (i.e., $K \sim c\|\Omega - \Omega^*\|^2$ in the relevant directions). Then the effective LLC near a stratum boundary satisfies:*

1. *For $\delta(\Omega) \gg \varepsilon^{1/2}$ (far from boundary): The LLC equals the interior value $\lambda_{S,\mathcal{B}}$.*

2. *For $\delta(\Omega) \sim \varepsilon^{1/2}$ (near boundary): Finite-sample LLC estimates show increased variance, scaling as $\mathrm{Var}(\hat{\lambda}) \sim \delta(\Omega)^{-1}$.*

3. *For $\delta(\Omega) \ll \varepsilon^{1/2}$ (very near boundary): The effective LLC interpolates between adjacent strata values.*

Proof sketch. Under local regularity, the sublevel set $\{K \leq \varepsilon\}$ has diameter $O(\varepsilon^{1/2})$ in the regular directions. When this diameter exceeds $\delta(\Omega)$, the sublevel set crosses stratum boundaries, mixing contributions from multiple strata.

Caveat: In singular directions where K vanishes faster than quadratically, the scaling $\varepsilon^{1/2}$ is replaced by $\varepsilon^{1/r}$ for the appropriate vanishing order r. The qualitative behavior (crossing threshold at comparable scales) persists. □

9.7 Combined degradation: A unified bound

We now combine all degradation effects into a unified bound.

Theorem 10 (Unified degradation bound)**.** *Consider a SubRep system with:*

- *Coupling where R may create independent singularities with LLC λ_R;*
- *Jacobian rank deficiency $r \geq 0$ in shared representation;*
- *$|\mathcal{A}|$ convex cells with at most τ tangencies;*
- *Boundary distance $\delta > 0$ from stratum boundaries.*

The additivity gap satisfies

$$|\Delta\lambda| \leq \underbrace{\lambda_R}_{\textit{coupling}} + \underbrace{\frac{r}{2}}_{\textit{intersection}} + \underbrace{(|\mathcal{A}| - 1 + \tau)\frac{d_W - 1}{2}}_{\textit{tropical}}. \tag{14}$$

When the remainder R is structurally subordinate (no new singularities), $\lambda_R = 0$. The boundary distance affects finite-sample estimates but not the asymptotic LLC.

Proof. The coupling contribution follows from Theorem 8. Intersection and tropical contributions follow from Propositions 9 and 10. The terms combine (approximately) additively since they arise from geometrically independent effects. □

Corollary 6 (Conditions for exact additivity)**.** *The additivity gap $\Delta\lambda = 0$ exactly when:*

1. *R is structurally subordinate ($\lambda_R = 0$);*
2. *$r = 0$ (transverse intersection);*
3. *$|\mathcal{A}| = 1$ (convex motive geometry);*
4. *$\tau = 0$ (no tangencies).*

9.8 Graceful vs. catastrophic degradation

The degradation bound (14) reveals two regimes:

Definition 23 (Degradation regimes)**.**

1. ***Graceful degradation:*** *$|\Delta\lambda|$ varies continuously as geometric parameters change (e.g., as singular varieties move transversely).*
2. ***Catastrophic degradation:*** *$|\Delta\lambda|$ jumps discontinuously (e.g., at stratum boundaries or when new singularities appear).*

Proposition 12 (Characterization of regimes)**.**

1. ***Within a stratum with subordinate coupling:*** *$\Delta\lambda$ is determined by geometric terms (intersection, tropical) and varies continuously.*
2. ***At stratum boundaries:*** *Catastrophic jumps by $|\lambda_{S_1,\mathcal{B}_1} - \lambda_{S_2,\mathcal{B}_2}|$ when crossing strata.*
3. ***When R creates new singularities:*** *Catastrophic if the new singularities have different vanishing order than K_0.*

Example 4 (Graceful regime). A SubRep system with:

- Structurally subordinate coupling ($\lambda_R = 0$);
- Full-rank Jacobian ($r = 0$);
- Nearly convex motive geometry ($|\mathcal{A}| = 2$, $\tau = 0$);
- Optimum in stratum interior.

Bound: $|\Delta\lambda| \leq 0 + 0 + (d_W - 1)/2 = (d_W - 1)/2$. For $d_W = 3$: $|\Delta\lambda| \leq 1$, a modest contribution from the single cell boundary.

Example 5 (Catastrophic regime). A SubRep system with:

- Coupling R that creates independent singularities ($\lambda_R = 3$);
- Rank-2 deficiency ($r = 2$);
- Highly non-convex geometry ($|\mathcal{A}| = 20$, $\tau = 5$).

Bound: $|\Delta\lambda| \leq 3 + 1 + 24 \cdot (d_W - 1)/2$. For $d_W = 5$: $|\Delta\lambda| \leq 3 + 1 + 48 = 52$. The non-convex geometry dominates, making modular decomposition essentially meaningless.

9.9 Empirical signatures of degradation

Remark 10 (Detecting degradation empirically). The following empirical signatures indicate departure from the graceful regime:

1. **LLC variance:** High variance in LLC estimates across training runs suggests proximity to stratum boundaries.
2. **LLC drift:** Monotonic drift in LLC during training suggests the system is traversing strata or approaching $\eta = 1$.
3. **Discrepancy:** Large gap between $\hat{\lambda}_{\text{goal}}$ and $\sum_k \hat{\lambda}_k$ directly measures $|\Delta\lambda|$.
4. **Sensitivity:** If small hyperparameter changes cause large LLC changes, the system is near a phase transition.

10 Beyond Additivity: LLC Interaction Information

The preceding sections treated non-additivity as deviation from an ideal. This section develops an alternative perspective: rather than viewing $\Delta\lambda$ as error, we interpret it as *meaningful structure* capturing how subgoal complexities interact. This leads to an inclusion-exclusion decomposition analogous to mutual information in information theory.

10.1 Motivation: From error to structure

In information theory, the joint entropy $H(X, Y)$ does not equal $H(X) + H(Y)$ in general. Rather than treating this as failure of additivity, Shannon introduced mutual information $I(X; Y) = H(X) + H(Y) - H(X, Y)$ as a fundamental quantity capturing statistical dependence.

We propose an analogous reframing for SLT: the joint LLC $\lambda(\theta_{\text{total}})$ generally differs from the sum of marginal LLCs $\sum_k \lambda(\theta_k)$, and this difference is not error but *interaction complexity*—a measure of how singular structures combine.

10.2 LLC interaction complexity: Two modules

Definition 24 (LLC interaction complexity)**.** *For two parameter blocks θ_A and θ_B with joint parameters $\theta_{AB} = (\theta_A, \theta_B)$, define the* LLC interaction complexity*:*

$$I_\lambda(A; B) := \lambda(\theta_A) + \lambda(\theta_B) - \lambda(\theta_{AB}), \tag{15}$$

where:

- *$\lambda(\theta_A)$ is the LLC of module A with B frozen at its optimum;*
- *$\lambda(\theta_B)$ is the LLC of module B with A frozen;*
- *$\lambda(\theta_{AB})$ is the LLC of the joint system.*

Remark 11 (Analogy to mutual information)*.* This mirrors the mutual information formula $I(X;Y) = H(X) + H(Y) - H(X,Y)$. However, a crucial difference is that I_λ can be *negative*, unlike Shannon mutual information.

10.3 Sign and interpretation of I_λ

The sign of $I_\lambda(A; B)$ has geometric meaning.

Proposition 13 (Interpretation of I_λ sign)**.**

1. *$I_\lambda(A; B) = 0$: Modules are* SLT-independent*. Their singular varieties do not interact; the joint singularity is the product of individual singularities.*
2. *$I_\lambda(A; B) > 0$:* Redundant complexity*. The modules share singular structure; jointly they are "simpler" than the sum of parts. This occurs when singular varieties intersect non-trivially.*
3. *$I_\lambda(A; B) < 0$:* Synergistic complexity*. The joint system has singularities that neither module has alone. This occurs when the coupling creates new singular directions.*

Proof. (1) Under exact factorization $K_{AB} = K_A + K_B$ with independent parameters, sublevel-set volumes convolve and LLCs add (Proposition 1), giving $I_\lambda = 0$.

(2) If modules share structure, the joint singular variety $\mathcal{S}_{AB} = \{K_{AB} = 0\}$ has higher codimension than expected from independent intersection: $\operatorname{codim}(\mathcal{S}_{AB}) > \operatorname{codim}(\mathcal{S}_A) + \operatorname{codim}(\mathcal{S}_B)$. By the LLC-codimension relationship, this gives $\lambda_{AB} < \lambda_A + \lambda_B$.

(3) Coupling can create singularities absent in either marginal. For example, if $K_A = \theta_A^2$, $K_B = \theta_B^2$, but $K_{AB} = (\theta_A - \theta_B)^2 + \epsilon(\theta_A^2 + \theta_B^2)$ for small ϵ, the joint system has a singular direction $\theta_A = \theta_B$ not present in either marginal. □

Example 6 (Shared representation induces $I_\lambda > 0$)*.* Consider options A and B sharing a feature extractor ψ. Each option's head has LLC $\lambda_A^{\text{head}}, \lambda_B^{\text{head}}$. If we compute "marginal" LLCs by freezing the other:

$$\lambda(\theta_A) = \lambda_\psi + \lambda_A^{\text{head}},$$
$$\lambda(\theta_B) = \lambda_\psi + \lambda_B^{\text{head}}.$$

But the joint system has $\lambda(\theta_{AB}) = \lambda_\psi + \lambda_A^{\text{head}} + \lambda_B^{\text{head}}$ (the shared ψ contributes once). Thus:

$$I_\lambda(A; B) = \lambda_\psi > 0.$$

The shared representation creates redundant complexity.

Example 7 (Coupling creates $I_\lambda < 0$). Let $K_A(\theta_A) = \theta_A^2$ and $K_B(\theta_B) = \theta_B^2$, both smooth with $\lambda_A = \lambda_B = 1/2$ (standard Gaussian).

Now suppose the joint loss has a coupling term: $K_{AB} = \theta_A^2 + \theta_B^2 + c(\theta_A\theta_B)^2$ for $c > 0$. The additional term creates a singularity along $\theta_A = 0$ or $\theta_B = 0$ axes that is "sharper" than either marginal. Computing (via resolution of singularities): $\lambda_{AB} > 1 = \lambda_A + \lambda_B$, so $I_\lambda(A; B) < 0$.

This represents synergistic complexity: the coupling creates singular structure absent in either module alone.

10.4 Inclusion-exclusion decomposition

For multiple modules, we obtain an inclusion-exclusion formula.

Definition 25 (Higher-order LLC interaction)**.** *For modules $\{1, \ldots, K\}$, define the k-th order interaction:*

$$I_\lambda(S) := \sum_{T \subseteq S} (-1)^{|S|-|T|} \lambda(\theta_T) \tag{16}$$

for any subset $S \subseteq \{1, \ldots, K\}$, where $\lambda(\theta_T)$ is the LLC of the subsystem containing modules in T.

This is the LLC analog of *interaction information* (also called *co-information*) in information theory.

Theorem 11 (Inclusion-exclusion for LLC)**.** *The total LLC decomposes as:*

$$\lambda_{\text{total}} = \sum_{k=1}^{K} \lambda_k - \sum_{i<j} I_\lambda(i; j) + \sum_{i<j<k} I_\lambda(i; j; k) - \cdots + (-1)^{K+1} I_\lambda(1; \ldots; K). \tag{17}$$

Equivalently, inverting:

$$\lambda_{\text{total}} = \sum_{\emptyset \neq S \subseteq \{1,\ldots,K\}} (-1)^{|S|+1} I_\lambda^{(|S|)}(S), \tag{18}$$

where $I_\lambda^{(1)}(k) := \lambda_k$ and higher orders are defined recursively.

Proof. This is the Möbius inversion formula applied to the lattice of subsets. Define $f(S) := \lambda(\theta_S)$. Then $I_\lambda(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} f(T)$ is the Möbius transform. Inverting: $f(S) = \sum_{T \subseteq S} I_\lambda(T)$, which for $S = \{1, \ldots, K\}$ gives (17). □

10.5 SubRep-specific interaction structure

For SubRep systems, the interaction terms have interpretable meanings.

Definition 26 (SubRep interaction decomposition)**.** *For a SubRep system with shared representation ψ, options $\{o_1, \ldots, o_n\}$, MDN, and planner, define:*

$$I_\lambda^{\text{rep}}(o_i; o_j) := \textit{interaction via shared } \psi, \tag{19}$$

$$I_\lambda^{\text{func}}(o_i; \text{plan}) := \textit{functional coupling interaction}, \tag{20}$$

$$I_\lambda^{\text{motive}}(o_i; o_j) := \textit{interaction via motive geometry}, \tag{21}$$

$$I_\lambda^{\text{trop}}(o_i; o_j) := \textit{tropical (cell boundary) interaction}. \tag{22}$$

Proposition 14 (Decomposition of pairwise interaction)**.** *For options o_i, o_j in a SubRep system:*

$$I_\lambda(o_i; o_j) = I_\lambda^{\text{rep}}(o_i; o_j) + I_\lambda^{\text{func}}(o_i; o_j) + I_\lambda^{\text{motive}}(o_i; o_j) + I_\lambda^{\text{trop}}(o_i; o_j). \tag{23}$$

Each term has a characteristic sign:

- $I_\lambda^{\text{rep}} \geq 0$ *always (shared structure $\Rightarrow$ redundancy).*
- I_λ^{func} *can have either sign (depends on coupling type).*
- $I_\lambda^{\text{motive}} \geq 0$ *typically (overlapping motive regions).*
- I_λ^{trop} *can have either sign (cell boundary effects).*

Proof sketch. Decompose the joint singular variety into contributions from each source of interaction. Shared representations always reduce joint complexity (same parameters, counted once vs. twice). Functional coupling can create or destroy singularities depending on whether compositions are transverse. Motive overlap typically creates redundancy. Tropical effects depend on cell boundary geometry. □

10.6 Three-way and higher interactions in SubRep

Higher-order interactions capture non-pairwise effects.

Definition 27 (Three-way LLC interaction)**.** *For modules A, B, C:*

$$I_\lambda(A; B; C) := I_\lambda(A; B) + I_\lambda(A; C) + I_\lambda(B; C) - I_\lambda(A; BC) - I_\lambda(B; AC) - I_\lambda(C; AB) + I_\lambda(A; B; C)_{\text{pure}}, \tag{24}$$

where $I_\lambda(A; B; C)_{\text{pure}}$ is the irreducible three-way contribution.

Proposition 15 (Interpretation of three-way interaction)**.**

1. $I_\lambda(A; B; C) > 0$*:* Redundant synergy*. The three-way combination reveals shared structure not visible in pairs.*
2. $I_\lambda(A; B; C) < 0$*:* Synergistic emergence*. The three-way combination creates singularities absent in any pair.*
3. $I_\lambda(A; B; C) = 0$*: Three-way effects reduce to pairwise.*

Example 8 (Three-option interaction in SubRep). Consider three options o_1, o_2, o_3 covering complementary aspects of a task (e.g., navigation, manipulation, communication for a robot).

If the planner composes them sequentially, each pair has functional coupling: $I_\lambda(o_i; o_j) < 0$ (synergy from composition).

But if all three together enable a capability none can achieve in pairs (e.g., coordinated behavior), then $I_\lambda(o_1; o_2; o_3) < 0$ represents emergent complexity from the triad.

Conversely, if all three share a common bottleneck (e.g., all require the same perceptual feature), then $I_\lambda(o_1; o_2; o_3) > 0$ represents triple-redundancy beyond pairwise sharing.

10.7 Partial information decomposition for LLC

Drawing further on information theory, we can decompose interactions into redundancy, synergy, and unique contributions.

Definition 28 (LLC-PID decomposition)**.** *For target T and source modules A, B, decompose:*

$$\lambda(T) = \mathrm{Uniq}_\lambda(T; A) + \mathrm{Uniq}_\lambda(T; B) \tag{25}$$
$$+ \mathrm{Red}_\lambda(T; A, B) + \mathrm{Syn}_\lambda(T; A, B), \tag{26}$$

where:

- $\mathrm{Uniq}_\lambda(T; A)$*: complexity of T attributable uniquely to A;*
- $\mathrm{Uniq}_\lambda(T; B)$*: complexity attributable uniquely to B;*
- $\mathrm{Red}_\lambda(T; A, B)$*: redundant complexity (shared by A and B);*
- $\mathrm{Syn}_\lambda(T; A, B)$*: synergistic complexity (arising only from A and B together).*

Proposition 16 (Constraints on LLC-PID)**.** *The LLC-PID satisfies:*

$$\lambda(T \mid A) = \mathrm{Uniq}_\lambda(T; B) + \mathrm{Syn}_\lambda(T; A, B), \tag{27}$$
$$\lambda(T \mid B) = \mathrm{Uniq}_\lambda(T; A) + \mathrm{Syn}_\lambda(T; A, B), \tag{28}$$
$$I_\lambda(A; B) = \mathrm{Red}_\lambda(T; A, B) - \mathrm{Syn}_\lambda(T; A, B), \tag{29}$$

where $\lambda(T \mid A) := \lambda(T, A) - \lambda(A)$ is the "conditional LLC."

Remark 12 (Operational meaning for SubRep)*.* In SubRep:

- $\mathrm{Uniq}_\lambda(\text{goal}; o_i)$: complexity contribution unique to option o_i.
- $\mathrm{Red}_\lambda(\text{goal}; o_i, o_j)$: complexity that either o_i or o_j could provide (substitutable options).
- $\mathrm{Syn}_\lambda(\text{goal}; o_i, o_j)$: complexity requiring both options together (complementary options).

This provides a principled basis for option library design: maximize unique and synergistic contributions, minimize redundancy.

Remark 13 (SLT advantage over Shannon PID)*.* In standard information theory, Partial Information Decomposition (PID) is notoriously ill-defined: there is no universally accepted way to decompose mutual information into unique, redundant, and synergistic components because the lattice of redundancy admits multiple consistent measures.

SLT may offer a *cleaner* PID than Shannon entropy. The reason is geometric: LLC is determined by intersection multiplicities of algebraic varieties (Proposition 14), and algebraic intersection theory provides unambiguous definitions of how varieties overlap. Unlike probability distributions, where "shared information" is ambiguous, singular varieties either intersect or they don't, and intersection multiplicities are well-defined integers.

Specifically:

- Red_λ corresponds to the LLC contribution from $\mathcal{S}_A \cap \mathcal{S}_B$ (shared singular locus).
- $\mathrm{Uniq}_\lambda(A)$ corresponds to the LLC from $\mathcal{S}_A \setminus \mathcal{S}_B$ (singular directions unique to A).
- Syn_λ corresponds to singularities in $\mathcal{S}_{AB}$ not present in $\mathcal{S}_A \cup \mathcal{S}_B$ (emergent singular directions).

The geometric definitions are unambiguous, suggesting SLT-PID may be better-behaved than Shannon-PID.

10.8 Interaction complexity and the additivity gap

We can now reinterpret the additivity gap.

Proposition 17 (Additivity gap as total interaction)**.** *The additivity gap equals the alternating sum of all interactions:*

$$\Delta\lambda = -\sum_{|S|\geq 2} (-1)^{|S|} I_\lambda(S) = -\sum_{i<j} I_\lambda(i;j) + \sum_{i<j<k} I_\lambda(i;j;k) - \cdots \tag{30}$$

Note the signs: positive pairwise interaction (redundancy, $I_\lambda > 0$) decreases $\Delta\lambda$*, while negative pairwise interaction (synergy, $I_\lambda < 0$)* increases $\Delta\lambda$.

Proof. Direct consequence of Theorem 11: $\Delta\lambda = \lambda_{\text{total}} - \sum_k \lambda_k$ and substituting the inclusion-exclusion expansion. □

Corollary 7 (Dominant pairwise approximation)**.** *If higher-order interactions are small relative to pairwise:*

$$\Delta\lambda \approx \sum_{i<j} I_\lambda(i;j). \tag{31}$$

This approximation is accurate when modules interact primarily in pairs.

10.9 Estimating interaction complexity

Remark 14 (Empirical estimation)*.* To estimate $I_\lambda(A;B)$ empirically:

1. Estimate $\hat{\lambda}_A$ by freezing B at its optimum and computing the LLC of A via standard methods (WBIC, evidence, etc.).
2. Estimate $\hat{\lambda}_B$ similarly.
3. Estimate $\hat{\lambda}_{AB}$ for the joint system.
4. Compute $\hat{I}_\lambda(A;B) = \hat{\lambda}_A + \hat{\lambda}_B - \hat{\lambda}_{AB}$.

Variance of $\hat{I}_\lambda$ combines variances of the three estimates.

Remark 15 (Computational cost)*.* For K modules, estimating all pairwise interactions requires $O(K^2)$ LLC estimates. Higher-order interactions require $O(2^K)$ estimates in general, but sparse interaction structures (e.g., tree-structured dependencies) can reduce this.

10.10 Geometric interpretation: Intersection multiplicities

The interaction complexity has a geometric interpretation in terms of intersection theory.

Proposition 18 (Interaction as intersection multiplicity)**.** *Let $\mathcal{S}_A, \mathcal{S}_B$ be the singular varieties of modules A, B. Then:*

$$I_\lambda(A;B) = \lambda_A + \lambda_B - \lambda_{AB} = \mu(\mathcal{S}_A \cap \mathcal{S}_B) - \mu_{\text{transverse}}, \tag{32}$$

where $\mu(\mathcal{S}_A \cap \mathcal{S}_B)$ is the intersection multiplicity and $\mu_{\text{transverse}}$ is the expected multiplicity under transverse intersection.

Proof sketch. Under transverse intersection, $\lambda_{AB} = \lambda_A + \lambda_B$ and $I_\lambda = 0$. Non-transverse intersection (tangency, containment) increases or decreases the intersection multiplicity, reflected in λ_{AB}. The difference measures departure from transversality. $\square$

Corollary 8 (Tropical interaction)**.** *For non-convex motive geometries, the tropical correction λ_{trop} contributes to interaction complexity:*

$$I_\lambda^{\text{trop}}(A;B) = \sum_{\sigma \in \mathcal{T}_A \cap \mathcal{T}_B} \mu(\sigma) \cdot \lambda_\sigma,$$

the sum over cells where tropical hypersurfaces of A and B intersect.

10.11 Design implications: Interaction-aware architecture

The interaction framework suggests new design principles.

1. **Minimize unintended redundancy.** Large positive $I_\lambda(o_i; o_j)$ indicates options o_i, o_j share structure unnecessarily. Consider factoring out shared components explicitly.

2. **Exploit intentional synergy.** Negative $I_\lambda(o_i; o_j)$ indicates options create emergent capability. Design option libraries to maximize beneficial synergies.

3. **Monitor interaction spectrum.** Track pairwise I_λ during training. Changes indicate restructuring of how options relate.

4. **Sparse interaction structure.** If most $I_\lambda(o_i; o_j) \approx 0$, the system is approximately modular. Target architectures with sparse interaction graphs.

5. **Bound higher-order interactions.** If pairwise interactions dominate, the inclusion-exclusion truncates early and complexity is predictable. Many-body interactions indicate highly entangled systems.

10.12 The synergy–stability tradeoff

There is a fundamental tension between the "synergy" of Section 10 and the "stability" of Section 9. This subsection makes the connection explicit.

Proposition 19 (Synergy–stability duality)**.** *Let $I_\lambda(A;B) < 0$ indicate synergistic complexity (emergent singularities from coupling). Then:*

1. *The coupling ratio η satisfies $\eta \geq |I_\lambda(A;B)|/(\lambda_A + \lambda_B)$.*

2. *If $|I_\lambda(A;B)| \geq \lambda_A + \lambda_B$, then $\eta \geq 1$ and the system is in the catastrophic degradation regime of Section 9.*

Proof. Synergistic complexity means $\lambda_{AB} > \lambda_A + \lambda_B$, i.e., the joint system has more singular structure than the sum of parts. This excess singularity arises from the coupling term $R(\theta)$. The coupling ratio $\eta = |R|/K_0$ must be large enough to create this additional singularity. Quantitatively, $\eta \geq (\lambda_{AB} - \lambda_A - \lambda_B)/(\lambda_A + \lambda_B) = |I_\lambda|/(\lambda_A + \lambda_B)$. $\square$

Remark 16 (Two faces of the same phenomenon). The "synergy" of Section 10 and the "catastrophic degradation" of Section 9 are *two perspectives on the same mathematical phenomenon*:

- **Capability perspective (Section 10):** Strong negative I_λ means emergent capabilities—the combined system can do things neither module can do alone.
- **Stability perspective (Section 9):** Strong coupling ($\eta \to 1$) means the additive approximation fails and LLC becomes unpredictable under perturbations.

Neither perspective is "correct"—they capture complementary aspects of the same underlying geometry.

Definition 29 (Synergy–stability frontier)**.** *For a family of architectures, the* synergy–stability frontier *is the Pareto-optimal set in the space of* (*total synergy*, *stability margin*)*, where:*

$$\text{Total synergy} := -\sum_{i<j} \min(I_\lambda(o_i; o_j), 0), \tag{33}$$

$$\text{Stability margin} := 1 - \eta. \tag{34}$$

Architectures on the frontier achieve maximal emergent capability for a given level of modularity (and vice versa).

Proposition 20 (Architectural tradeoffs on the frontier)**.**

1. ***Fully modular:*** *$I_\lambda \approx 0$ for all pairs. High stability ($\eta \approx 0$), zero synergy. The system is predictable but has no emergent capabilities.*
2. ***Fully entangled:*** *Large negative I_λ for all pairs. High synergy, low stability ($\eta \to 1$). The system has strong emergent capabilities but is fragile and hard to analyze.*
3. ***Hierarchical structure:*** *Cluster options into groups with high within-group synergy and low between-group interaction. This achieves local emergence with global modularity.*

Remark 17 (Design heuristic)*.* A practical design heuristic is to *maximize local synergy while minimizing global synergy*:

- **Within a subgoal cluster:** Allow strong coupling ($I_\lambda \ll 0$) to enable emergent capabilities.
- **Between clusters:** Enforce weak coupling ($I_\lambda \approx 0$) to preserve modularity and trainability.

This hierarchical structure places the system on the synergy–stability frontier with favorable properties: emergent capabilities within functional units, predictable composition across units.

Example 9 (Robot task decomposition)*.* Consider a robot with options for navigation, manipulation, and communication.

- **Within navigation:** Options for path-planning, obstacle avoidance, and localization may have $I_\lambda < 0$ (synergy)—together they enable robust navigation neither achieves alone.
- **Between navigation and manipulation:** Ideally $I_\lambda \approx 0$—the navigation module's complexity should be independent of whether the robot is carrying an object.
- **Pathological case:** If navigation and manipulation have $I_\lambda \ll 0$, changes to manipulation affect navigation unpredictably. The system is powerful but fragile.

11 Implications for SubRep Design

Our theoretical development suggests several principles for SubRep system design and diagnostics.

11.1 Architectural conditions for tractable decomposition

Theorem 5 identifies conditions ensuring LLC additivity:

1. **Accurate option models.** Low option prediction error ensures dominated coupling (Proposition 2).
2. **Smooth planner.** Lipschitz planner objectives reduce coupling strength.
3. **Option separation.** Options should address distinct task aspects, minimizing cross-gradients.
4. **Non-degenerate representations.** The shared feature extractor should have full-rank Jacobian.
5. **Simple motive geometry.** Fewer convex cells and transverse cell boundaries minimize tropical corrections.

11.2 Motive geometry design principles

The non-convex analysis (Section 7) suggests:

1. **Prefer convex or nearly-convex geometries.** If the motive space can be approximated by a single convex cone, $\lambda_{\text{trop}} = 0$ and the analysis simplifies.
2. **Minimize cell count.** Use the coarsest convex decomposition consistent with task requirements. Fine decompositions increase λ_{trop} and stratification complexity.
3. **Avoid degenerate cell boundaries.** Ensure cell boundaries intersect transversely (generically). Tangent or high-multiplicity intersections create larger tropical corrections.
4. **Keep optima away from boundaries.** If Ω^* is far from cell transition boundaries (in the interior of a double stratum), $\lambda_{\text{trop}} \approx 0$ effectively.
5. **Use soft cell transitions.** The soft relaxation (Proposition 7) smooths cell boundaries, reducing singularities during training.

11.3 Diagnostic signatures

The SLT framework provides diagnostic tools:

1. **Stability detection.** Stable LLC values (estimated empirically) indicate the system is in a consistent decompositional regime within a single double stratum.
2. **Option phase transitions.** Sudden LLC changes when options are admitted or removed suggest stratum boundary crossings.
3. **Motive phase transitions.** LLC changes without option changes suggest motive cell transitions—the active region of the motive geometry has shifted.
4. **Coupling detection.** If $\lambda_{\text{goal}} \gg \sum_o \lambda_o + \lambda_\psi + \ldots$, coupling terms are significant, and the factorization assumptions should be revisited.
5. **Tropical signature.** Large λ_{trop} (estimated via proximity to cell boundaries) suggests the motive geometry is too finely decomposed or has degenerate structure.

11.4 Training with non-convex geometries

1. **Temperature annealing.** Begin training with high temperature (soft cell transitions) and anneal toward hard boundaries. This smooths the loss landscape early and allows the optimizer to find good basins before encountering cell boundary singularities.

2. **Cell-aware initialization.** Initialize parameters so Ω_0 is in the interior of a single cell, avoiding immediate exposure to cell boundary singularities.

3. **Adaptive cell refinement.** Start with a coarse convex decomposition and refine only where needed. Monitor LLC to detect when refinement causes problematic singularities.

4. **Regularization toward convexity.** Add regularization terms encouraging the MDN to produce more convex motive geometries, trading expressiveness for simpler LLC structure.

11.5 Soft vs. hard admission tradeoffs

Our analysis suggests practical benefits to soft CDS/PDS:

- **Gradient flow.** Soft margins allow gradients to flow through admission decisions, improving trainability.

- **Smooth LLC landscape.** The LLC varies continuously with parameters, avoiding jumps.

- **Annealing.** Training can begin with high temperature (soft) and anneal toward hard decisions, tracking LLC evolution.

11.6 Limitations and extensions

Our analysis has limitations that suggest directions for future work:

1. **Finite-sample estimation.** We have characterized population quantities; practical LLC estimation requires careful finite-sample analysis, including variance bounds and convergence rates for empirical LLC estimators.

2. **Infinite cell decompositions.** Our non-convex analysis assumes finitely many cells. Continuous parameterizations of motive geometry (e.g., smooth manifolds) require extending the tropical framework to infinite-dimensional settings.

3. **Dynamic option libraries.** Our stratification is static; online option discovery with continuous option addition/removal requires a dynamic stratification theory with moving boundaries.

4. **Explicit computation.** Computing LLCs for specific SubRep architectures (e.g., with neural network components) requires resolving singularities of neural network loss landscapes, which remains challenging.

5. **Higher-order tropical corrections.** Our tropical LLC analysis captures leading-order contributions. Higher-order corrections from complex cell boundary intersections (triple points, etc.) may be significant in highly non-convex geometries.

6. **Non-semi-algebraic geometries.** Some MDN outputs may produce non-semi-algebraic motive geometries (e.g., involving transcendental functions). Extension to o-minimal structures may be necessary.

12 Conclusion

We have developed a rigorous SLT analysis of SubRep systems, addressing functional coupling, shared representations, non-analytic CDS/PDS boundaries, non-convex motive geometries, robustness of additivity, and—most significantly—moving beyond additivity to an interaction-theoretic framework. Our main contributions are:

1. **Coupling analysis.** We identified sufficient conditions (option accuracy, planner smoothness, option separation) under which functional coupling remains dominated, preserving approximate LLC additivity (Proposition 2, Lemma 1).

2. **Shared representation theory.** We showed how shared feature extractors contribute their LLC once (not per-module) to the total, with an intersection correction for non-transverse singular varieties (Theorem 1).

3. **Stratified SLT.** We introduced soft CDS/PDS relaxations and proved existence of a well-defined stratified LLC for hard systems, characterizing discontinuities at stratum boundaries (Theorem 2, Proposition 3).

4. **Non-convex motive geometries.** We developed a complete theory for non-convex motive spaces via convex decomposition, double stratification over options and cells, and tropical geometry. The tropical LLC correction λ_{trop} quantifies singularities from piecewise motive structure (Theorem 3, Theorem 4, Proposition 8).

5. **Robustness and degradation analysis.** We characterized when the additivity gap $\Delta\lambda$ is zero (structurally subordinate coupling, transverse intersection, convex geometry) versus when it is nonzero (Theorems 7, 8, 10). We distinguished graceful degradation (continuous variation) from catastrophic failure (discontinuous jumps at stratum boundaries or when new singularities appear) (Proposition 12).

6. **LLC interaction complexity.** We introduced $I_\lambda(A;B) = \lambda_A + \lambda_B - \lambda_{AB}$ as the SLT analog of mutual information, with positive values indicating redundant complexity (shared singular structure) and negative values indicating synergistic complexity (emergent singularities). The inclusion-exclusion formula (Theorem 11) decomposes total LLC via pairwise, three-way, and higher-order interactions, reinterpreting the additivity gap as total interaction rather than error. We also identified the *synergy–stability tradeoff* (Section 10.12): synergistic complexity ($I_\lambda < 0$) enables emergent capabilities but is mathematically equivalent to the strong coupling that causes catastrophic degradation. This duality suggests designing hierarchical architectures with local synergy and global modularity.

7. **Main decomposition theorem.** We combined these elements into a comprehensive decomposition result (Theorem 5) with robust bounds even when coupling is strong (Theorem 6).

The interaction-theoretic perspective (Section 10) represents a conceptual shift: rather than asking "when does additivity hold?" we ask "what is the interaction structure?" This reframing has several advantages:

- Non-additivity becomes *informative* rather than problematic. $I_\lambda(o_i; o_j) > 0$ reveals shared structure; $I_\lambda < 0$ reveals emergent capabilities.

- The inclusion-exclusion formula provides a *complete* decomposition, not an approximation with error bounds.

- Design guidance shifts from "minimize coupling" to "design beneficial interactions"—maximize synergy within functional clusters, minimize coupling between clusters.
- The geometric foundation of SLT may yield cleaner PID than Shannon information theory, since intersection multiplicities are unambiguous.
- The synergy–stability frontier provides a framework for understanding architectural tradeoffs between capability and predictability.

For practical SubRep systems, our results suggest:

1. **Additivity as baseline.** When interactions are small ($|I_\lambda| \ll \lambda_k$), the additive approximation suffices.
2. **Pairwise correction.** When higher-order interactions are small, $\lambda_{\text{total}} \approx \sum_k \lambda_k - \sum_{i<j} I_\lambda(i;j)$ captures most structure.
3. **Full interaction analysis.** For highly entangled systems, the complete inclusion-exclusion is necessary.
4. **Interaction-aware design.** Monitor I_λ during training to detect option redundancy (split or merge options) and synergy (preserve beneficial couplings).

These results provide a principled foundation for understanding complexity in hierarchical goal decomposition systems, moving from a naive additive model to a rich interaction-theoretic framework that embraces the non-linear combination of subgoal complexities as meaningful structure.

A Proof Details

A.1 Proof of Lemma 1 (full)

We provide the complete argument for the volume bound under dominated coupling.

Proof. Let $K_0(\Omega) := \sum_k K_k(\theta^{(k)})$ denote the sum of module excess losses. Under Assumption 1(a):

$$(1-\eta)K_0(\Omega) \le K(\Omega) \le (1+\eta)K_0(\Omega) \tag{35}$$

for all $\Omega \in U$.

Upper bound on $V(\varepsilon)$:

From the left inequality in (35):

$$\{K(\Omega) \le \varepsilon\} \subseteq \{K_0(\Omega) \le \varepsilon/(1-\eta)\}.$$

For the sum $K_0 = \sum_k K_k$, we use the convolution structure. Define $\varepsilon' := \varepsilon/(1-\eta)$. Then:

$$\begin{aligned} V_{K_0}(\varepsilon') &:= \mathrm{Vol}\{K_0 \le \varepsilon'\} \\ &= \int_{\sum_k \varepsilon_k \le \varepsilon'} \prod_k V_k'(\varepsilon_k)\, d\varepsilon_1 \cdots d\varepsilon_{K-1}, \end{aligned}$$

where $V_k'(\varepsilon) := dV_k/d\varepsilon$.

For each $V_k(\varepsilon) \sim c_k \varepsilon^{\lambda_k} |\log \varepsilon|^{m_k - 1}$, we have $V_k'(\varepsilon) \sim c_k \lambda_k \varepsilon^{\lambda_k - 1} |\log \varepsilon|^{m_k - 1}$.

The convolution integral yields (by standard asymptotic analysis):

$$V_{K_0}(\varepsilon') \sim C\varepsilon'^{\sum_k \lambda_k} |\log \varepsilon'|^{\sum_k (m_k-1)+K-1}.$$

Up to multiplicative constants and logarithmic corrections, this establishes $V(\varepsilon) \le c_+ \prod_k V_k(\varepsilon')$.

Lower bound on $V(\varepsilon)$:

From the right inequality in (35):

$$\{K_0(\Omega) \le \varepsilon/(1+\eta)\} \subseteq \{K(\Omega) \le \varepsilon\}.$$

The same convolution analysis gives $V(\varepsilon) \ge c_- \prod_k V_k(\varepsilon'')$ for $\varepsilon'' := \varepsilon/(1+\eta)$.

Prior factorization:

Under Assumption 1(c), the prior contributes bounded multiplicative factors c_1, c_2 that absorb into c_-, c_+. □

A.2 LLC additivity via Laplace-Stieltjes transform

For completeness, we sketch why LLCs add under independence, using the correct sublevel-set argument (not direct zeta factorization).

Proof of Proposition 1. With $K(\theta) = \sum_k K_k(\theta^{(k)})$ and $\phi(\theta) = \prod_k \phi_k(\theta^{(k)})$:

$$\begin{aligned} \zeta(s) &= \int_\Theta K(\theta)^s \phi(\theta)\, d\theta \\ &= \int_\Theta \Big(\sum_k K_k(\theta^{(k)})\Big)^s \prod_k \phi_k(\theta^{(k)})\, d\theta. \end{aligned}$$

This does *not* directly factorize due to the power of a sum. However, we use the Mellin transform representation. For a single block:

$$\zeta_k(s) := \int_{\Theta_k} K_k(\theta^{(k)})^s \phi_k(\theta^{(k)})\, d\theta^{(k)}.$$

The pole structure of ζ_k at $s = -\lambda_k$ determines the volume scaling.

For independent blocks, the total volume satisfies $V(\varepsilon) = \mathrm{Vol}\{\sum_k K_k \le \varepsilon\}$. The Laplace-Stieltjes transform of V factors:

$$\int_0^\infty e^{-t\varepsilon} dV(\varepsilon) = \prod_k \int_0^\infty e^{-t\varepsilon_k} dV_k(\varepsilon_k).$$

Tauberian theorems relate the transform's behavior as $t \to \infty$ to $V(\varepsilon)$ as $\varepsilon \to 0$, yielding $\lambda = \sum_k \lambda_k$. □

B Comparison with Previous Work

Our treatment extends previous theoretical notes on SubRep-SLT connections in several ways:

1. **Coupling.** Previous work assumed exact factorization or asserted dominated interaction without proof. We provide explicit conditions (Proposition 2) under which SubRep architectures satisfy these conditions.

2. **Shared parameters.** Previous work assumed disjoint parameter blocks. Theorem 1 handles shared representations, showing the shared block contributes once.

3. **Non-analyticity.** Previous work noted LLC "jumps" without rigorous foundation. Theorem 2 and Proposition 3 provide the stratified analysis justifying these phenomena.

4. **Non-convex geometries.** Previous work assumed convex motive cones. Section 7 develops the complete theory for non-convex geometries via convex decomposition, double stratification, and tropical geometry, with explicit bounds on the tropical correction (Proposition 8).

5. **Robustness analysis.** Previous work gave no indication of how additivity degrades when assumptions weaken. Section 9 provides quantitative bounds on the additivity gap (Theorem 10), characterizes graceful versus catastrophic degradation regimes (Proposition 12), and identifies the phase transition at $\eta = 1$ (Theorem 9).

6. **Beyond additivity.** Previous work treated non-additivity as error to be bounded. Section 10 reframes non-additivity as *meaningful structure* via LLC interaction complexity I_λ, the SLT analog of mutual information. The inclusion-exclusion decomposition (Theorem 11) provides a complete characterization rather than error bounds. This connects SLT to information-theoretic concepts (mutual information, interaction information, partial information decomposition) not previously explored.

7. **Proofs.** Previous work asserted results without proof. We provide complete proofs (some relegated to Appendix A).